HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY AND EDUCATION

Faculty of International Education

TELECOMMUNICATIONS

DESIGN THE AIR QUALITY MONITORING SYSTEM USING ESP8266 AND BLYNK SERVER MONITORING

Lecturer: **Ph.D. Phạm Ngọc Sơn**

Students : **Võ Hoàng Khang**

ID: *21161245*

Nguyễn Phạm Mai Phương

ID: *21161260*

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Table of contents

CHAPTER 1: INTRODUCTION **1**

I. Problem Statement **1**

II. Description of the Project **4**

III. Benefits of the system **4**

IV. Objectives **5**

V. Project plan **5**

VI. Estimate cost **7**

CHAPTER 2: LITERATURE REVIEW AND THEORETICAL FRAMEWORK **8**

I. Literature review **8**

1. MQTT Based Air Quality Monitoring System using NodeMCU and Node-RED 8

2. Design of Internet of Things Based Air Pollution Monitoring System Using ThingSpeak and Blynk Application 9

II. Theoretical Framework **10**

1. The Oled 10

1.2. Pin 10

1.3. Parameter 11

2. Temperature & humidity sensor (DHT11) 11

2.2. Pin 11

2.3. Parameter 12

3. Air quality sensor (MQ135) 12

3.2. Pin 13

3.3. Parameter: 13

4. Module Wifi ESP8266 13

4.1. Introduction 13

4.2. Pins 15

4.3. Parameter 15

CHAPTER 3: DESIGN AND IMPLEMENTATION **16**

I. PROJECT REQUIREMENTS **16**

1. Hardware Requirements: 16

2. Software Requirements: 16

II. SYSTEM DESIGN 16
1. Block diagram 16
2. Hardware design 18
2.1. Calculation for choosing devices 18
2.2. Hardware circuit design and connection 20
3. Software design 20

CHAPTER 4: EXPERIMENTS AND DESIGN ANALYSIS 25

I. Design Experiments 25
1. Software design: 25
2. Hardware design: 26

II. Operation Experiment 27

III. Design Analysis 29
1. Checking delay 29
2. The accuracy of the sensors 29
3. Sensor range testing 31

IV. Result Analysis 32

CHAPTER 5: RESULTS 34

CHAPTER 6: CONCLUSION AND RECOMMENDATION 36

I. Conclusion of the entire project 36

II. Suggestions for future project development 36

REFERENCES 37

APPENDIX A 38

APPENDIX B 44

LIST OF FIGURE

Fig. 1.1. Disease burden by risk factor, world, 2019 1 1

Fig. 1.2. Death rate from air pollution, Viet Nam, 2016 to 2019 1 2

Fig. 1.3. Share of deaths attributed to to air pollution, 2016 to 2019 1 3

Fig. 2.1. I2C OLED 0,91 inch 128x321 10

Fig. 2.2. Temperature and Humidity Sensor 1 11

Fig. 2.3. Air quality sensor MQ-135 and the air quality index table 1 13

Fig. 2.4. ESP8266 1 14

Fig. 2.5. Pins diagram of ESP8266 1 15

Fig. 3.1. Block Diagram of whole system 1 17

Fig. 3.2. Block Diagram of Sensor Node 1 17

Fig. 3.3. Fritzing system connection1 20

Fig. 3.4. Main Flowchart1 21

Fig. 3.5. Flowchart for Debouncing Button 1 22

Fig. 3.6. sendSensor() Flowchart1 23

Fig. 3.7. setup() Flowchart1 24

Fig. 4.1. Image of the IDE programming window and the test board circuit 1 25

Fig. 4.2. Schematic and PCB design on Proteus 1 26

Fig. 4.3. Image of the printed circuit board and the completed sensor node product 1..... 27

Fig. 4.4. Image of monitoring the Sensor Node displaying temperature and humidity 1 27

Fig. 4.5. Image of monitoring the Sensor Node displaying air quality 1..... 28

Fig. 4.6. Image of monitoring warning at sensor node 1 28

Fig. 4.7. Chart for evaluating technical 1..... 33

Fig. 5.1. The final image of the sensor node from the project 1..... 34

CHAPTER 1: INTRODUCTION

I. Problem Statement

The decision to focus on the design and construction of an air quality monitoring system using ESP8266 through a smartphone stems from the growing concern over air pollution globally and its impact on public health, the environment, and the economy. The urgency of addressing air quality issues is highlighted by numerous studies and data indicating the adverse effects of air pollutants on human health and the environment.

The World Health Organization (WHO) estimates that about 7 million people die prematurely each year from air pollution-related diseases, making it one of the leading environmental health risks worldwide.

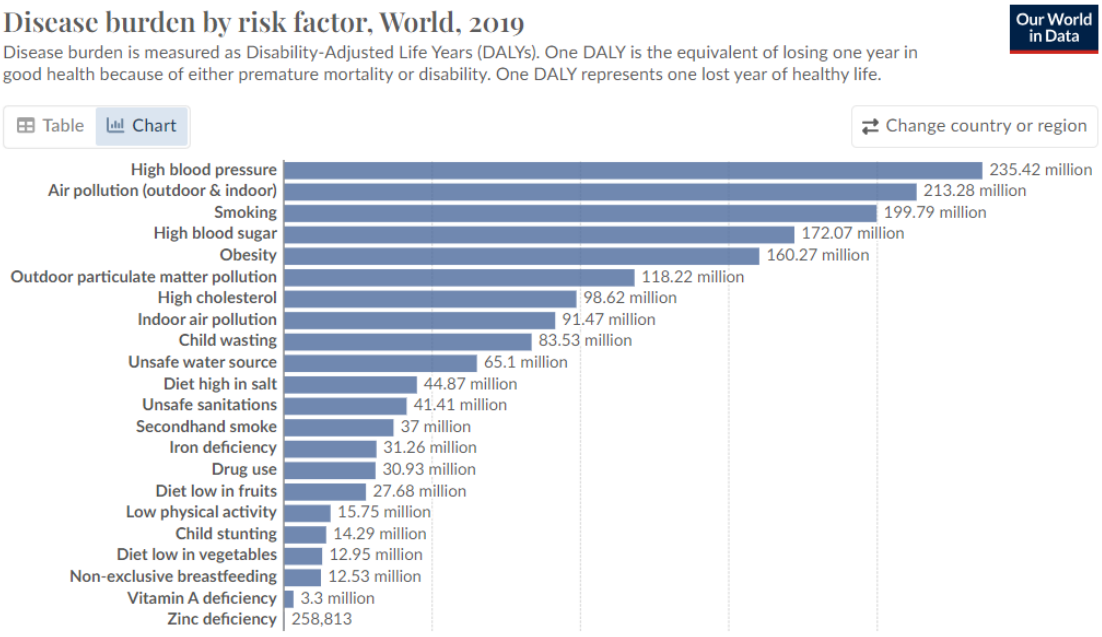


Fig. 1.1. Disease burden by risk factor, world, 2019 [1]

In Vietnam, rapid urbanization and industrialization have exacerbated air pollution, especially in major cities like Hanoi and Ho Chi Minh City. These urban areas frequently

report Air Quality Index (AQI) levels that surpass safe thresholds, posing significant health risks to the population.

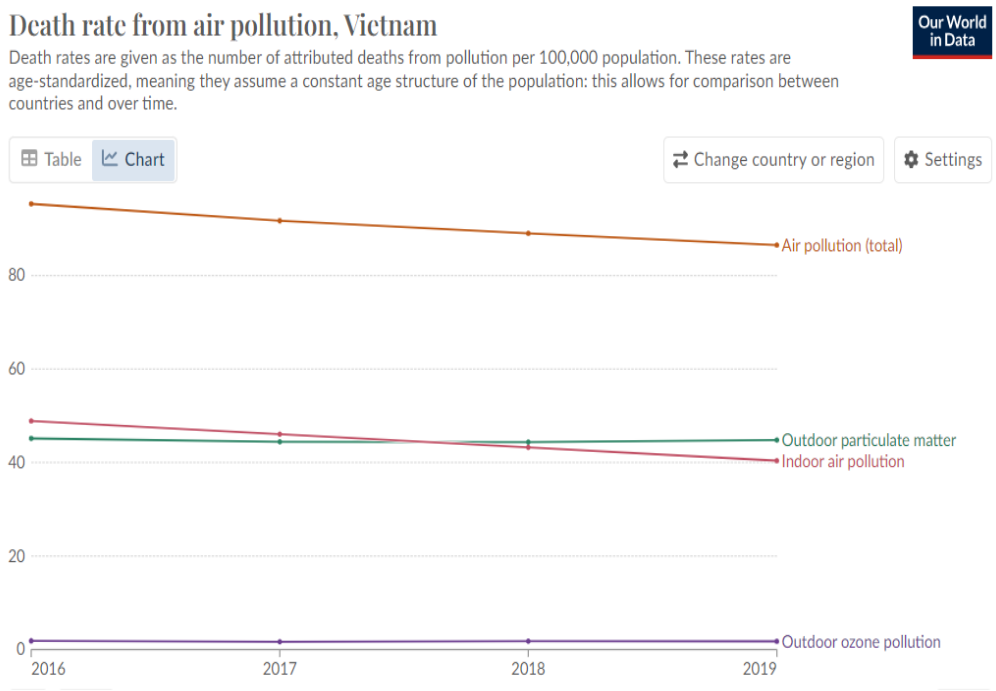


Fig. 1.2. Death rate from air pollution, Viet Nam, 2016 to 2019 [1]

The chart reflects the situation in Vietnam where urbanization and industrialization have led to increased levels of air pollution, particularly from outdoor sources. This trend aligns with the observation that cities like Hanoi and Ho Chi Minh City often experience unsafe AQI levels, contributing to health risks and the overall burden of disease from air pollution.

The mortality rates due to air pollution are notably higher in countries with low to middle incomes, and while air pollution is a global health and environmental concern, the severity of its impacts varies widely across different regions.

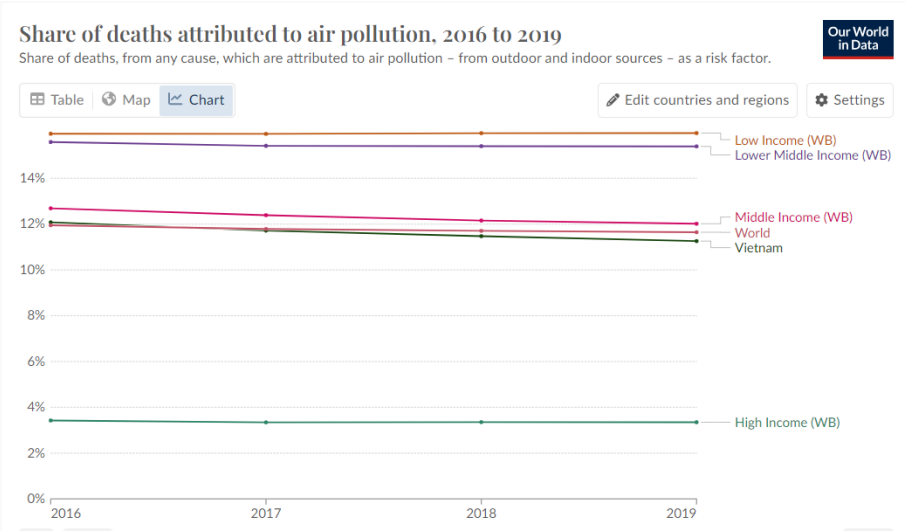


Fig. 1.3. Share of deaths attributed to to air pollution, 2016 to 2019 *1*

The line chart indicates that the high rates of air pollution-related health issues in low-to-middle-income countries, including Vietnam, indicate a clear need for affordable air quality monitoring devices. Access to such technology would empower individuals and communities to understand their exposure to pollutants and advocate for necessary environmental health measures.

Despite the significant impact of air pollution, there remains a gap in affordable, accessible, and accurate air quality monitoring systems. Existing solutions often involve sophisticated and expensive equipment that is not readily available or accessible to the general public.

Therefore, many solutions are considered to tackle this problem. One of the most measure is using Wireless Sensor Networks (WSN) which has gained significance in diverse applications in monitoring. Wireless sensor network system worn to monitor and control the air quality of an environment is developed. The air pollution monitoring system that measures temperature, humidity and CO are proposed.

Analytical measuring paraphernalia is lavish, time and power overriding, and can seldom be used for air quality exposure in real time. Our project research and design a device which use IOT and wireless sensor networks to be cost-effective.

II. Description of the Project

Our project aims to monitor indoor air quality, focusing on temperature, humidity, carbon dioxide (CO2) levels, and gas detection. The system utilizes affordable and widely available sensors to ensure cost-effectiveness without compromising functionality.

The main sensors used in the project are the DHT11 and MQ-135. The DHT11 sensor is chosen for its affordability, availability, and suitability for monitoring indoor temperature and humidity. With its appropriate temperature and humidity measurement range, it is ideal for indoor environments. Additionally, the MQ-135 sensor is integrated to detect air quality parameters such as CO2 concentration and other gases.

To process sensor data and support remote monitoring, the ESP8266 microcontroller is chosen as the main processor. Its low cost, adequate processing power, and sufficient GPIO pins make it a suitable choice for this application. The ESP8266 can connect to Wi-Fi networks, allowing communication with external devices such as smartphones for remote monitoring.

Furthermore, to provide real-time feedback to users, an OLED display is integrated with the system. The OLED display serves as a local sensor parameter display interface, saving energy. Additionally, sensor data is transmitted to the user's smartphone for remote monitoring via Wi-Fi connection. This feature enables users to monitor indoor air quality from anywhere, providing convenience and peace of mind.

III. Benefits of the system

Air quality monitoring system brings many benefits to your health and living environment.

First and foremost, this is a product that helps people identify pollution levels, thereby taking appropriate health protection measures, especially for sensitive people such as children, the elderly, and people with respiratory diseases. In addition, it warns

of potential risks from toxic substances such as dust PM2.5, CO2 gas. That helps you prevent health from problems such as cough, asthma, allergies, and even cancer.

Then, this product provides information for you to adjust your living, working, and studying environment to become healthier, helping you feel comfortable and improve work efficiency.

Moreover, this device helps you monitor air quality changes over time, thereby evaluating the effectiveness of environmental improvement measures and providing data for research on air pollution, contributing to research on air pollution

IV. Objectives

The objective of the project is to design an air quality monitoring system using ESP8266 and Blynk server. It aims to integrate sensors for measuring air quality, temperature, and humidity with ESP8266 microcontroller. The system will provide real-time monitoring locally via an OLED display and remotely through the Blynk IoT platform. Users will receive alerts and notifications regarding significant changes in air quality levels. The project also focuses on enhancing the user interface, ensuring system stability, and providing comprehensive documentation and support.

V. Project plan

PLANNING	ASSIGNED TO	PROGRESS	START	END
Determine the project title and the reasons for selecting the topic.	All Team	100%	26/2/2024	3/3/2024
Define the implementation plan and brainstorm ideas.	All Team	100%	26/2/2024	3/3/2024

Research the technical aspects and calculate the estimated costs.	All Team	100%	26/2/2024	10/3/2024
Evaluate and consider product criteria.	M.Phương	100%	26/2/2024	10/3/2024
DESIGN	**ASSIGNED TO**	**PROGRESS**	**START**	**END**
Studying theory, programming button inputs, basic LED control and how to upload for the ESP8266.	All Team	100%	3/3/2024	10/3/2024
Studying the theory and programming techniques for OLED display.	H.Khang	100%	3/3/2024	24/3/2024
Studying the theory of programming and data calculation for the MQ-135 sensor.	H.Khang	100%	3/3/2024	24/3/2024
Studying the theory of programming and data calculation for the temperature and humidity sensor.	M.Phương	100%	3/3/2024	24/3/2024
Proceeding with the conceptual design of the hardware circuit for the sensor node.	All Team	100%	24/3/2024	31/3/2024

Proceeding with programming and uploading the firmware to the hardware.	All Team	100%	31/3/2024	7/4/2024
Programming the remote monitoring link via the phone.	All Team	100%	7/4/2024	14/4/2024
EVALUATION	**ASSIGNED TO**	**PROGRESS**	**START**	**END**
Test, experiment, and make necessary adjustments.	All Team	100%	14/4/2024	28/4/2024

VI. Estimate cost

The estimated cost for a single sensor node is as follows:

- ESP8266 CP2101: 78,000 VND
- DHT11 Sensor: 15,000 VND
- MQ-135 Sensor: 32,000 VND
- 0.91 inch I2C OLED Display: 52,000 VND
- Resistors, LEDs, buttons, PCB fabrication, etc.: 100,000 VND

Total Estimated Cost: 277,000 VND

Total Estimated Cost in USD: Approximately 11.70 USD

This cost estimation includes all the necessary components to build a fully functional sensor node for monitoring air quality, which can communicate with the Blynk server for data visualization and remote monitoring.

CHAPTER 2: LITERATURE REVIEW AND THEORETICAL FRAMEWORK

I. Literature review

1. MQTT Based Air Quality Monitoring System using NodeMCU and Node-RED

In **[1]** Somphop Chanthakit & Choopan Rattanapoka described the implementation of an MQTT-based air quality monitoring system using an ESP8266 NodeMCU. The device measures temperature, humidity, CO, O3, and PM2.5 levels. It acts as a publisher, sending sensor data to an MQTT Broker. Node-RED, acting as a subscriber, processes this data and displays it on a responsive web dashboard with gauges, text, and charts. If any measurements exceed set thresholds, Node-RED sends alarm notifications via LINE Notify.

Advantages:

- **Low Cost:** Utilizes affordable components such as the ESP8266 NodeMCU, DHT22, MQ-131, MQ-7 and Dust Sensor, making the system cost-effective.
- **Remote Monitoring:** Enables remote observation of air quality data through an MQTT server, accessible via a responsive web application.
- **Notifications:** Provides alerts through LINE Notify, sending notifications to smartphones when sensor values exceed predefined thresholds.
- **User-Friendly Interface:** Displays data in gauge, text, and chart formats, making it easy to understand and manage the received data.

Disadvantages:

- **No Real-Time Monitoring on Mobile and at Sensor Node:** Does not support real-time monitoring directly from a smartphone, and also for the Sensor Node because it didn’t have any hardware for monitoring at the Node.

- **Limited Sensor Range:** The accuracy and range of the sensors might not be sufficient for all applications, particularly in environments with rapidly changing air quality conditions.

2. Design of Internet of Things Based Air Pollution Monitoring System Using ThingSpeak and Blynk Application

In **[2]**, the creation of an IoT-based air pollution monitoring system that measures carbon dioxide, butane gas, humidity, and temperature were presented by S.A.Z. Murad, F.A. Bakar, A. Azizan & M.A.M Shukri. The hardware includes an MQ-2 gas sensor, ESP8266 Wi-Fi module, and DHT22 sensors. The software is developed using the Arduino IDE. The system indicates air quality with an AQI scale: below 100 for safe and above 200 for hazardous. A green LED signals safe air, while a red LED warns of hazardous gases. Data is sent to ThingSpeak and Blynk, updating every 15 seconds and 1 second respectively. Blynk also sends immediate alerts when hazardous gas is detected.

Advantages:

- **Low Cost:** The system is designed to be affordable, making it accessible for widespread use.
- **Real-time Monitoring:** Users can monitor air quality in real-time via both mobile applications and web interfaces.
- **User-Friendly:** The system is easy to use with clear indicators (LEDs) and a simple interface for data visualization.
- **Notifications:** Immediate alerts are sent to users when air quality deteriorates, ensuring timely awareness and action.

Disadvantages:

- **Lack of On-Site Display:** There is no provision for displaying sensor data directly at the sensor node, requiring users to rely on remote monitoring.

- **Additional Costs:** The inclusion of the STM32 module introduces unnecessary costs that could be avoided with simpler alternatives.
- **Sensor Accuracy:** The sensors used may have limitations in accuracy, potentially affecting the reliability of the data**.**

II. Theoretical Framework

1. The Oled

1.1. Introduction to Oled

Oled LCD screen 0.91 ich with I2C interface for beautiful, luxurious display, clear daytime clarity and maximum energy saving with appropriate level, the screen uses I2C communication next for quality. Stable transmission quality and extremely easy communication with only 2-pin GPIO.[1]

Fig. 2.1. I2C OLED 0,91 inch 128x32[1]

1.2. Pin

- VCC: 2.2~5.5VDC
- GND: 0VDC
- SCL: Clock
- SDA: input data

[1] HSHOP, Màn Hình Oled 0.91 Inch Giao Tiếp I2C, https://hshop.vn/products/man-hinh-oled-0-91-inch-giao-tiep-i2c

1.3. Parameter

- Supplied Voltage: 2.2~5.5VDC
- Power Consumption: 0.03w
- Display angle: greater than 160 degrees
- Number of pixel displayed: 128x32 pixels
- Communication standard: I2C
- Driver: SSD1306

2. Temperature & humidity sensor (DHT11)

2.1. Introduction to DHT11

DHT11 is a temperature and humidity sensor that communicates with 1 data pin. DHT11 can measure humidity values from 20% to 90%RH and temperature from 0oC to 50oC, accuracy: ± 5%RH and ±2oC[2]

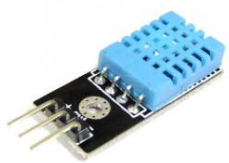


Fig. 2.2. Temperature and Humidity Sensor [1]

2.2. Pin

- **+:** VCC acceptable 3.5-5v.
- **- :** Ground connect with GND.
- **out:** Output data of temperature and humidity.

[2] Điện tử tương lai, DHT11- Temperature & Humidity sensor, https://dientutuonglai.com/cam-bien-nhiet-do-va-do-am-dht11.html

2.3. Parameter

- Operating voltage: 3V 5VDC
- Communication standard: TTL, 1 wire.
- Humidity measurement range: 20%-90% RH (error 5%RH)
- Temperature measurement range: 0-50°C (error 2°C)
- Maximum sampling frequency: 1Hz (1 second/time)
- Dimensions: 28mm x 12mm x10mm

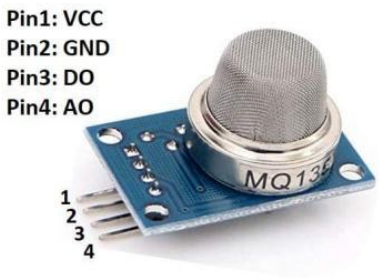
3. Air quality sensor (MQ135)

3.1. Introduction to MQ135

MQ135 Gas Sensor module for Air Quality having Digital as well as Analog output. Sensitive material of MQ135 gas sensor is SnO2, which with lower conductivity in clean air. When the target combustible gas exist, The sensors conductivity is more higher along with the gas concentration rising. MQ135 gas sensor has high sensitivity to Ammonia, Sulphide and Benze steam, also sensitive to smoke and other harmful gases. It is with low cost and suitable for different application.[3]

Used for family, Surrounding environment noxious gas detection device, Apply to ammonia, aromatics, sulfur, benzene vapor, and other harmful gases/smoke, gas detection, tested concentration range: 10 to 1000 ppm.

[3] MECSU, MQ-135 – "Gas sensor for air quality'', 1/08/2023. Available: https://mecsu.vn/ho-tro-ky-thuat/mq135-cam-bien-khi-cho-chat-luong-khong-khi.maw



Air Quality Index Levels of Health Concern	Numerical Value	Meaning
Good	0-50	Air quality is considered satisfactory, and air pollution poses little or no risk.
Moderate	51-100	Air quality is acceptable; however, for some pollutants there may be a moderate health concern for a very small number of people who are unusually sensitive to air pollution.
Unhealthy for Sensitive Groups	101-150	Members of sensitive groups may experience health effects. The general public is not likely to be affected.
Unhealthy	151-200	Everyone may begin to experience health effects; members of sensitive groups may experience more serious health effects.
Very Unhealthy	201-300	Health alert: everyone may experience more serious health effects.
Hazardous	> 300	Health warnings of emergency conditions. The entire population is more likely to be affected.

Fig. 2.3. Air quality sensor MQ-135 and the air quality index table [1]

3.2. Pin

- Pin1: It is VCC Pin. It is used to connect 2.5-5V to the sensor.
- Pin2: It is GND Pin. It is used to connect GND to the sensor.
- Pin3: It is digital output Pin. From this pin you will get digital data HIGH/LOW.
- Pin4: It is analog output Pin. From this pin you will get analog data.

3.3. Parameter:

- Power voltage: 5V DC
- Voltage of heater: 5V±0.1 AC/DC
- Load resistance: variable (2kΩ-47kΩ)
- Heater resistance: 33Ω±5%
- Power consumption of heater: less than 800mW
- Detection range: 10 - 300 ppm NH3, 10 - 1000 ppm Benzene, 10 - 300 Alcohol
- Dimension: 32mm*20mm

4. Module Wifi ESP8266

4.1. Introduction

The ESP8266 is a System on Chip (SoC) or microcontroller because it integrates a Tensilica L106 32-bit controller (MCU) and additionally integrates a Wi-Fi transceiver.

The ESP8266 microcontroller includes 17 GPIO pins, but only 11 of these GPIO pins are typically used as normal inputs/outputs, while the remaining 6 pins are used for code uploading and interfacing with the Flash memory chip on the circuit board. The Wi-Fi transceiver used on the ESP8266, combined with simple TCP/IP connections, makes it very useful for wireless Wi-Fi network applications. Thanks to this, you can connect web servers to real web pages to connect to smartphones. The company manufacturing this microcontroller is Espressif, based in China.[4]

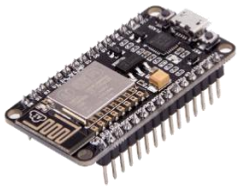


Fig. 2.4. ESP8266 *1*

Introduction to ESP8266 NodeMCU:

NodeMCU (Node MicroController Unit) is an open-source software and hardware development environment built around an affordable System on Chip (SoC) called ESP8266, which integrates multiple functionalities such as PWM, ADC reading, I2C communication, on the same module. Additionally, the board provides the ability to connect to a computer via microUSB standard, making it easy for developers to implement applications on this product.[5]

[4] Electronics Curiosities, "ESP8266 Node MCU board specifications and pins descriptions. Frequently asked questions related to WI-FI board esp8266 node MCU", 1/8/2021, Available: https://www.electronicscuriosities.com/2021/08/esp8266-node-mcu-board-specifications.html

[5] Make-It.ca, "NodeMCU ESP8266 Detailed Review", Available: https://www.make-it.ca/nodemcu-details-specifications/

4.2. Pins

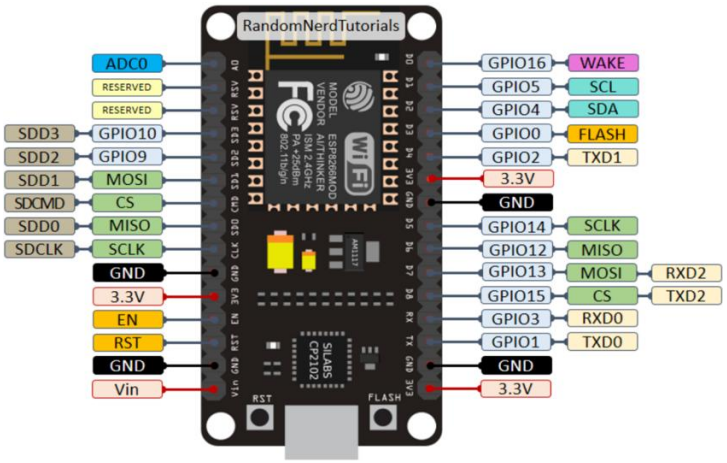


Fig. 2.5. Pins diagram of ESP8266 [1]

4.3. Parameter

- Microcontroller: ESP8266EX
- Wifi: 2.4 GHz supports 802.11 b/g/n
- Operating Voltage: 5V via micro USB port
- Input Voltage: 3.3V
- Number of I/O Pins: 11 (all I/O pins have Interrupt/PWM/I2C/One-wire, except D0)
- Flash Memory: 4 MB
- Communication: Micro USB Cable
- Security Support: WPA/WPA2
- Integrated Protocol: TCP/IP
- Dimensions: 49x26 mm

CHAPTER 3: DESIGN AND IMPLEMENTATION

I. Project requirements

1. Hardware Requirements:

The Hardware Requirements for this project include the following components:

- **ESP8266:** This serves as the main microcontroller for data processing and communication.
- **I2C OLED 0.91 inch:** A display unit used to visualize the collected data.
- **DHT11:** A humidity and temperature sensor for environmental monitoring.
- **MQ-135:** A gas sensor used for air quality monitoring.
- **Additional components** such as resistors, buttons, LEDs, and other small electronic components may also be required for circuit interfacing and control functionalities.

2. Software Requirements:

The Software Requirements for this project include:

- **Arduino IDE:** Used for programming and uploading code to the ESP8266 microcontroller.
- **Blynk:** A platform for connecting and sending data to the Blynk server, enabling real-time monitoring and control of the system from both smartphones and computers.
- **Proteus and Fritzing:** Designing circuits and drawing real circuits with Fritzing makes it easier to describe the circuit, while Proteus is used for designing schematic circuits and PCBs.

II. System Design

1. Block diagram

The basic system consists of Sensor Nodes connected to send data to the Blynk Server, allowing computers and phones to observe as follows:

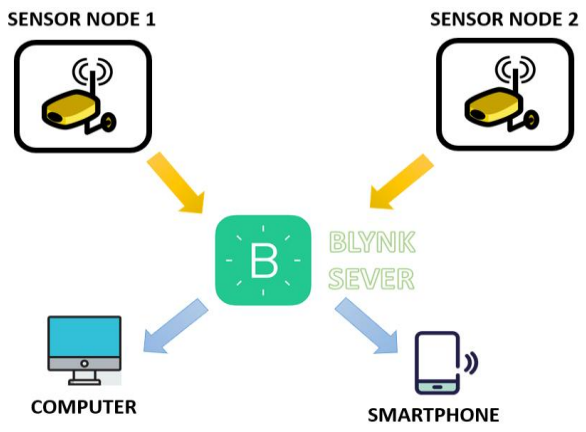


Fig. 3.1. Block Diagram of whole system *1*

The basic system block diagram consists of two Sensor Nodes connected to the Blynk Server, allowing them to transmit data. From the Blynk Server, both computers and smartphones can observe and monitor the data collected by the Sensor Nodes. This setup enables remote monitoring and access to real-time environmental information captured by the Sensor Nodes, enhancing the overall visibility and control of the system.

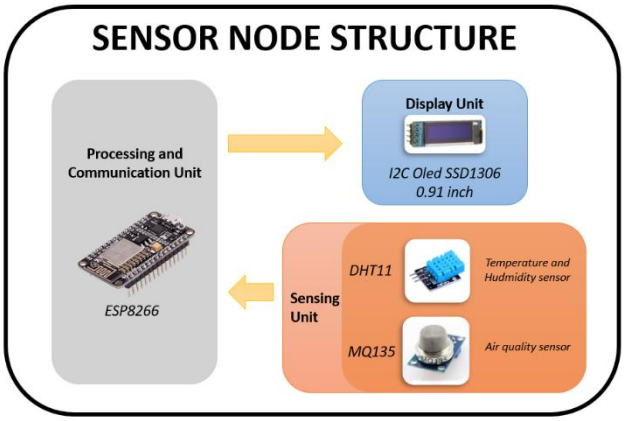


Fig. 3.2. Block Diagram of Sensor Node *1*

The sensor node block diagram consists of an ESP8266 module as the processing unit. The output of the processing unit is connected to a display unit, which is an I2C OLED 0.91 display. The input to the processing unit comes from the sensing unit, which comprises two main sensors: the DHT11 sensor for measuring temperature and humidity, and the

MQ135 sensor for measuring air quality. The ESP8266 module gathers data from these sensors, processes it, and then displays the relevant information on the OLED display in real-time. This setup enables continuous monitoring of environmental parameters such as temperature, humidity, and air quality.

2. Hardware design

2.1. Calculation for choosing devices

Firstly, we choose DHT11 to measure temperature and humidity of the air because some below reasons:

- **Cost-effective:** DHT11 sensors are very inexpensive, making them ideal for hobbyist projects or low-cost devices.
- **Easy to use:** DHT11 uses a simple digital protocol for communication, making it easy to integrate with the ESP8266.
- **Provides temperature and humidity:** This device needs both temperature and humidity for a basic understanding of air quality.

Then, we choose MQ135 to measure air quality. This is reasons why we choose it:

Why MQ-135 complements ESP8266:

- **Microcontroller capabilities:** The ESP8266 can handle analog or digital inputs from the MQ-135 and can be programmed to set thresholds for low air quality. When the sensor reading exceeds the threshold, the ESP8266 can trigger the indicator light.
- **Selectivity:** The MQ-135 might be sensitive to various gases, not just pollutants. Depending on the environment, it might require calibration or additional filtering in the code to focus on specific air quality concerns.
- **Accuracy:** MQ-135 offers moderate accuracy, suitable for basic air quality monitoring. For highly precise measurements, professional-grade sensors might be needed.

- **Baseline calibration:** MQ-135 sensors often require a pre-heating time and might need baseline calibration for optimal performance.

Choosing the I2C OLED 0.91-inch display offers several advantages:

- **Affordable Price:** The display is cost-effective, fitting within budget constraints for projects.
- **Minimal Pin Usage:** It requires fewer pins for communication, leaving more GPIO pins available for other components or functionalities.
- **Low Power Consumption: The** OLED display consumes minimal power, making it suitable for battery-operated or energy-efficient applications.

The ESP8266 (Arduino) is very well-suited for this device that uses DHT11 and MQ135 sensors for several reasons:

ESP8266 Capabilities:

- **Microcontroller:** The ESP8266 is a microcontroller, which means it's a tiny computer on a chip. It can read data from sensors (DHT11 and MQ135), process that information, and control outputs (indicator light) based on your program.
- **Digital and Analog Input:** The ESP8266 can handle both digital and analog inputs. DHT11 uses a digital protocol, while MQ135 can be found in either analog or digital output versions. The ESP8266 can work with both types.
- **Wi-Fi connectivity:** A key feature of the ESP8266 is built-in Wi-Fi connectivity. While not directly used in the scenario you described, it opens up possibilities for future enhancements. You could program the ESP8266 to send air quality data to your phone or the cloud for remote monitoring.

Suitability for DHT11 and MQ135:

- **Simple Communication Protocols:** Both DHT11 and MQ135 (depending on the version) use well-understood communication protocols. The ESP8266 can be programmed to understand these protocols and read sensor data easily.
- **Processing Power:** The ESP8266 has enough processing power to handle basic tasks like reading sensor data, comparing it to thresholds, and controlling the indicator light.
- **Cost-effective:** The ESP8266 is a relatively inexpensive microcontroller board, making it a good choice for hobbyist projects or cost-sensitive applications

2.2. Hardware circuit design and connection

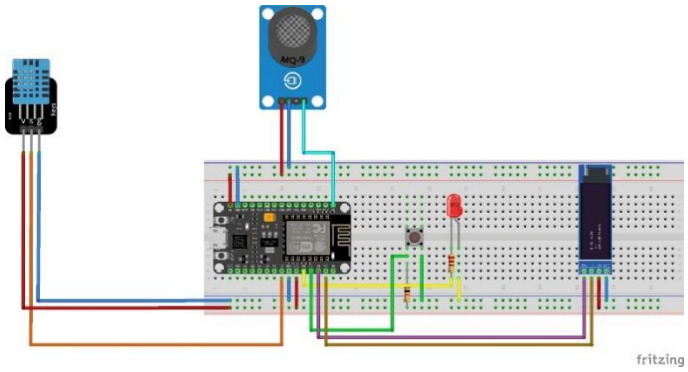


Fig. 3.3. Fritzing system connection[1]

The actual circuit is drawn using Fritzing, depicting a test board plug-in design as shown in the image above. The hardware circuit design involves connecting an ESP8266 module to an OLED display via I2C, along with DHT11 and MQ-135 sensors for temperature, humidity, and air quality measurement. GPIO pins on the ESP8266 are configured for sensor inputs and display output. Components are assembled on a breadboard for prototyping.

3. Software design

- **Main:**

The main program and the main() function serve to declare variables, libraries, configure pins, and call the required sub-functions within main(). It checks button variables and displays temperature, humidity, air quality sequentially, links with Blynk, and also monitors the MQ-135 ppm value to send events to Blynk.

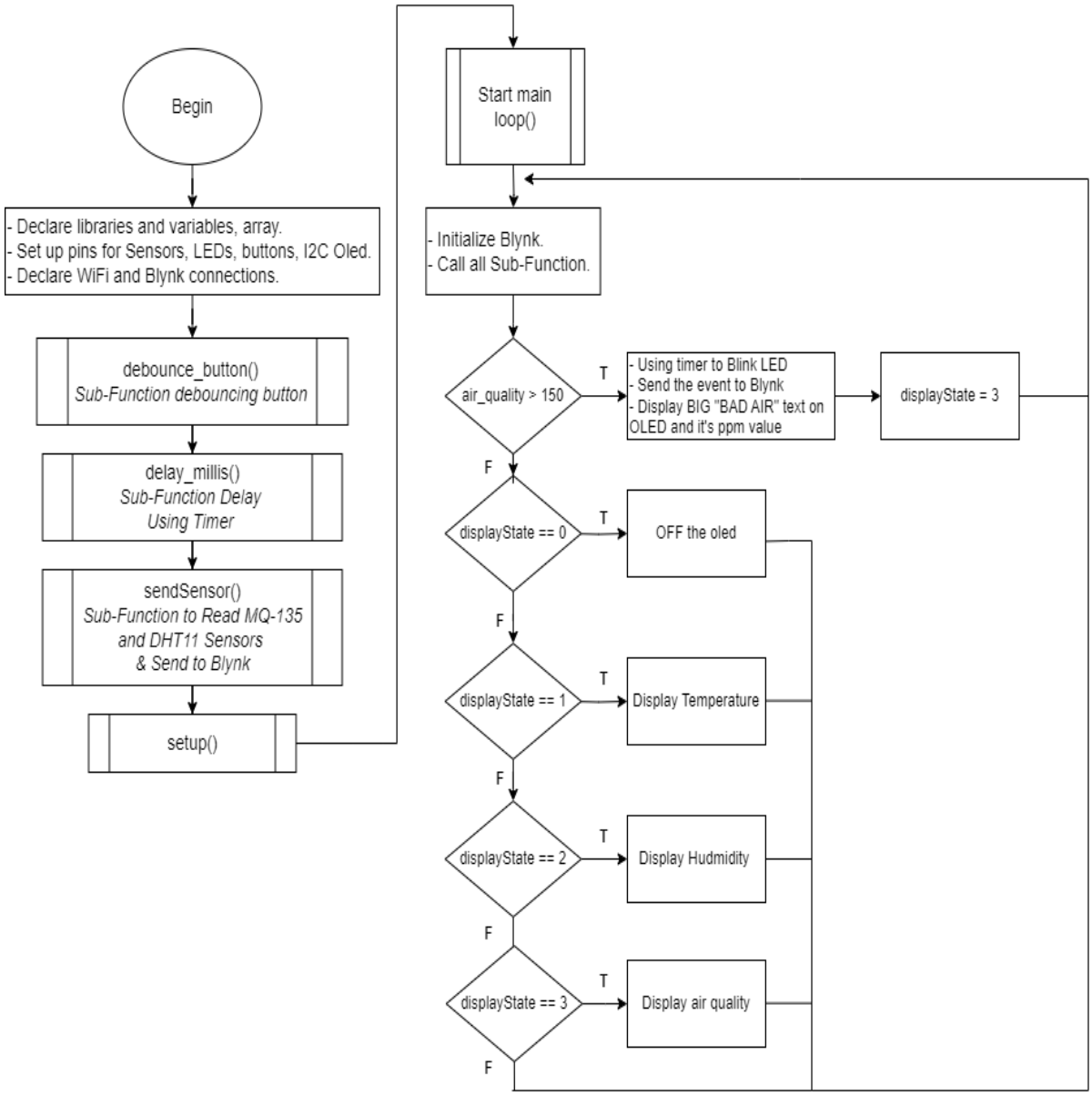


Fig. 3.4. Main Flowchart1

- **debounce_button():**

The debounce_button() sub-function serves to debounce the button press. It introduces a 20ms delay to prevent bouncing and waits for the button to settle. It also includes a display variable to control the display within the main() function.

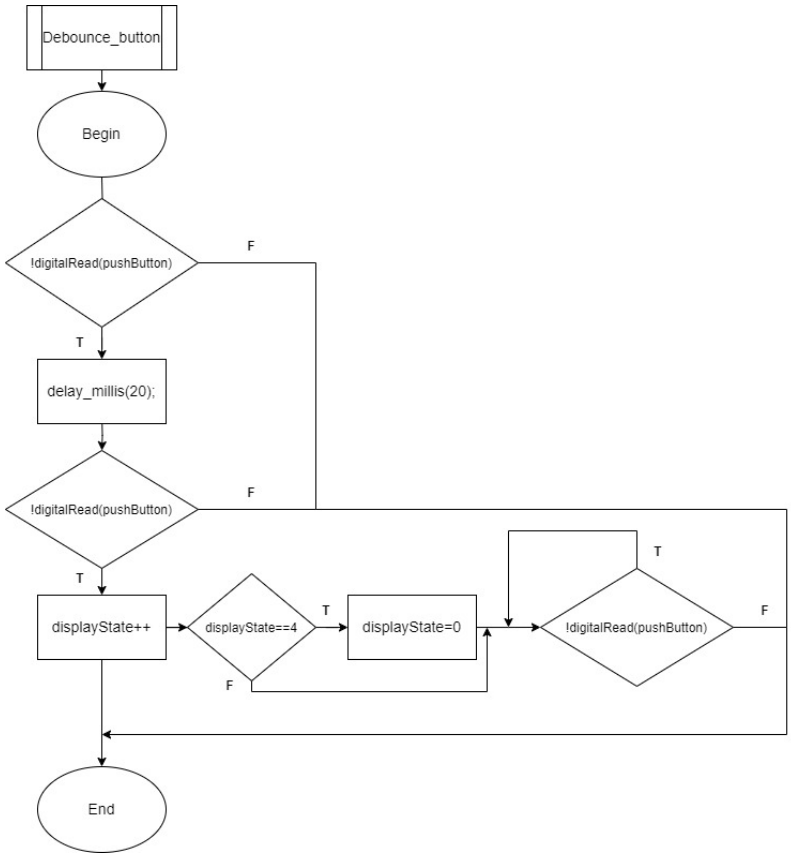


Fig. 3.5. Flowchart for Debouncing Button [1]

- **sendSensor():**

The sendSensor() sub-function is used to initialize the sensors, read their values, and link them with the predefined values on the Blynk Server.

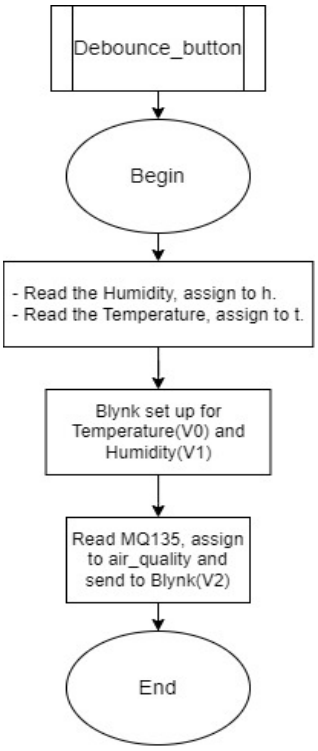


Fig. 3.6. sendSensor() Flowchart[1]

- **setup():**

The setup() function initializes values such as baud rate, pins for the button, LED, display variables, and OLED initialization. It also connects to WiFi and waits for a successful connection. Once connected, it displays the connection status on the OLED and lights up a warning LED.

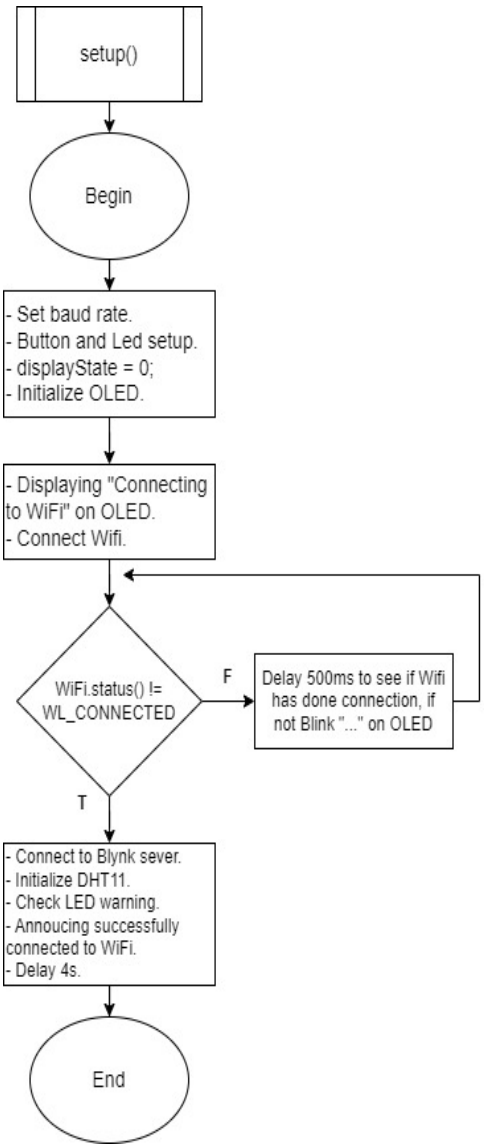


Fig. 3.7. setup() Flowchart[1]

CHAPTER 4: EXPERIMENTS AND DESIGN ANALYSIS

I. Design Experiments

1. Software design:

The software design for the air quality monitoring system involves programming the ESP8266 using the Arduino IDE. The code integrates sensor data from DHT11 and MQ-135, sends this data to the Blynk server, and displays it on an OLED screen. Initially, the program is developed and tested on a breadboard like in Fig 4.1 to ensure proper functionality and sensor accuracy. This setup allows for easy modifications and debugging before final deployment.

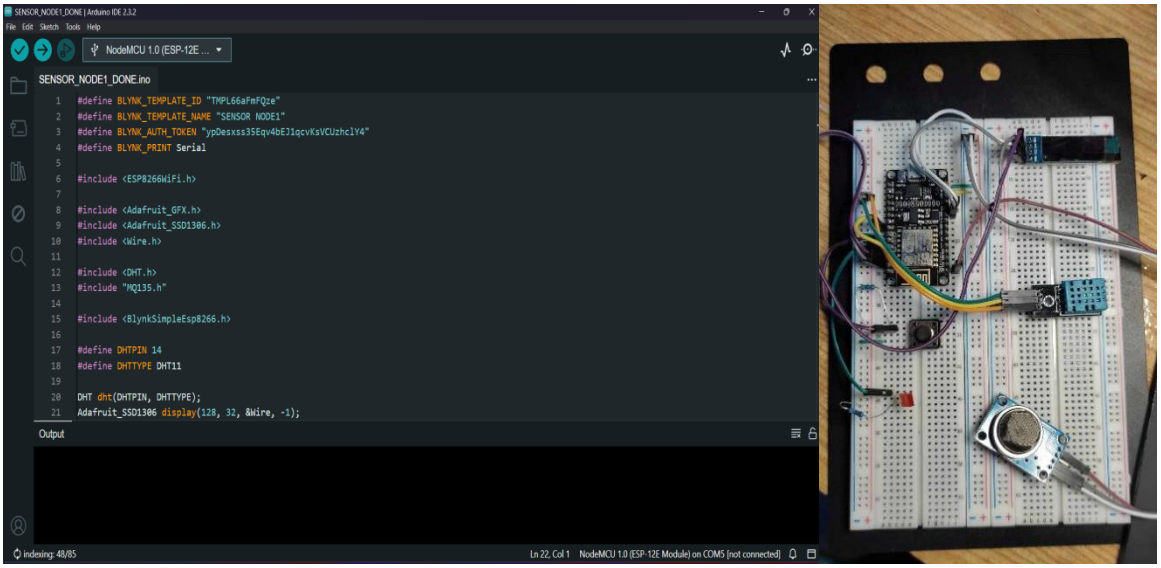


Fig. 4.1. Image of the IDE programming window and the test board circuit [1]

Additionally, the team also established a connection with the Blynk server to verify the sensor values being transmitted and to demonstrate the capability of remote monitoring via phone or computer. The complete code, explanations, and instructions for connecting to Blynk are detailed in Appendices A and B.

2. Hardware design:

After completing the software part, the team proceeded to design a PCB with the intention of providing convenient plug-in outputs for the sensors, OLED, buttons, LEDs, etc., as depicted in the diagram below.

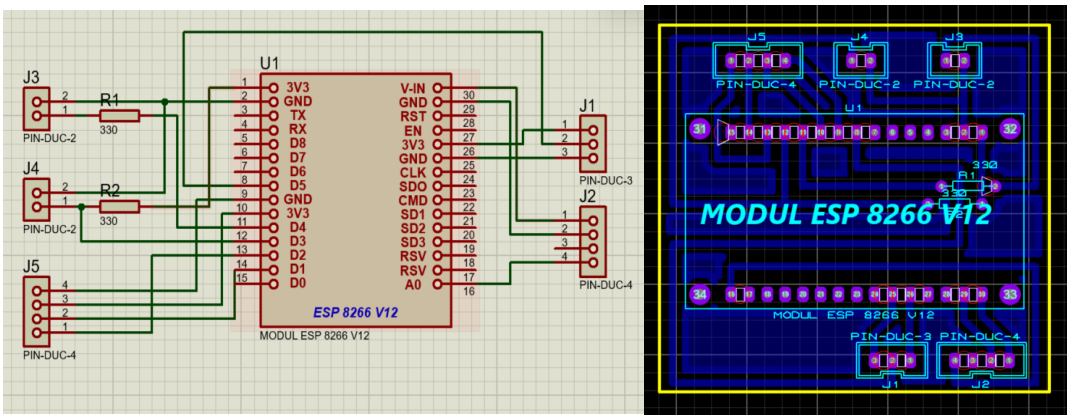


Fig. 4.2. Schematic and PCB design on Proteus [1]

Subsequently, the team proceeded to fabricate the printed circuit board (PCB) and solder the components, thereby completing the product.

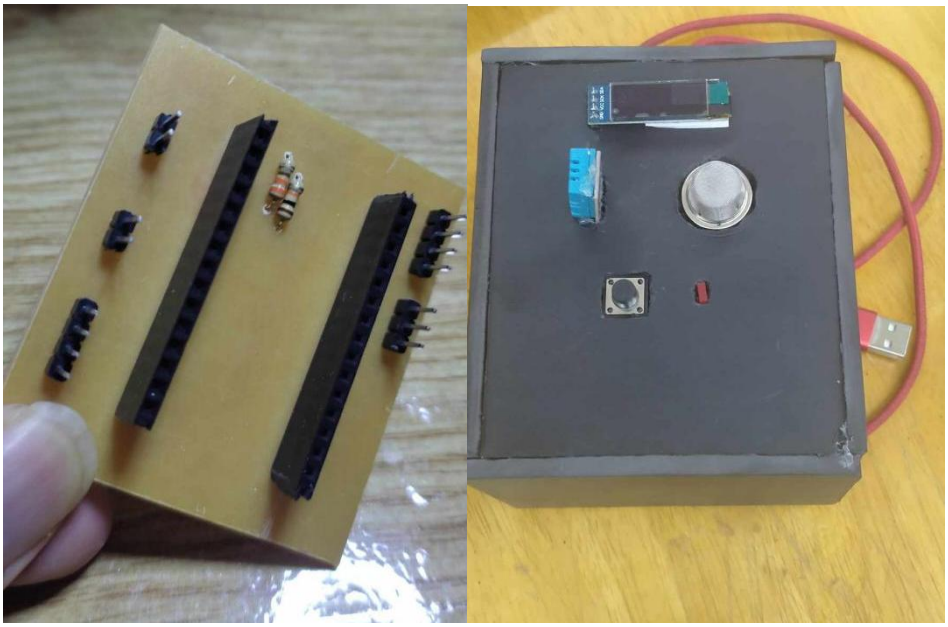


Fig. 4.3. Image of the printed circuit b oard and the completed sensor node product [1]

II. Operation Experiment

Following that is the operational validation of the Sensor Node's functionality, along with the observation link on Blynk.

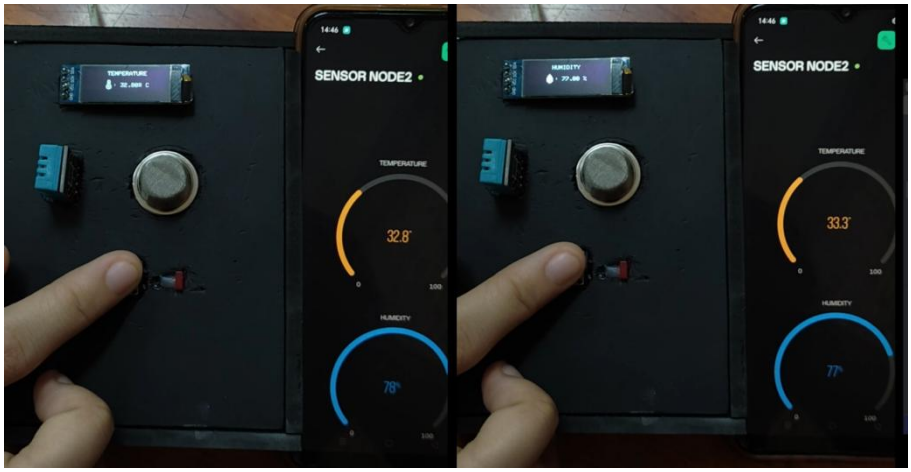


Fig. 4.4. Image of monitoring the Sensor Node displaying temperature and humidity [1]

Fig. 4.5. Image of monitoring the Sensor Node displaying air quality [1]

At Sensor Node 1, the operation is also successful, similar to Sensor Node 2. Next, the team proceeds to verify the alerting capability of the Sensor Node, as shown in the image below:

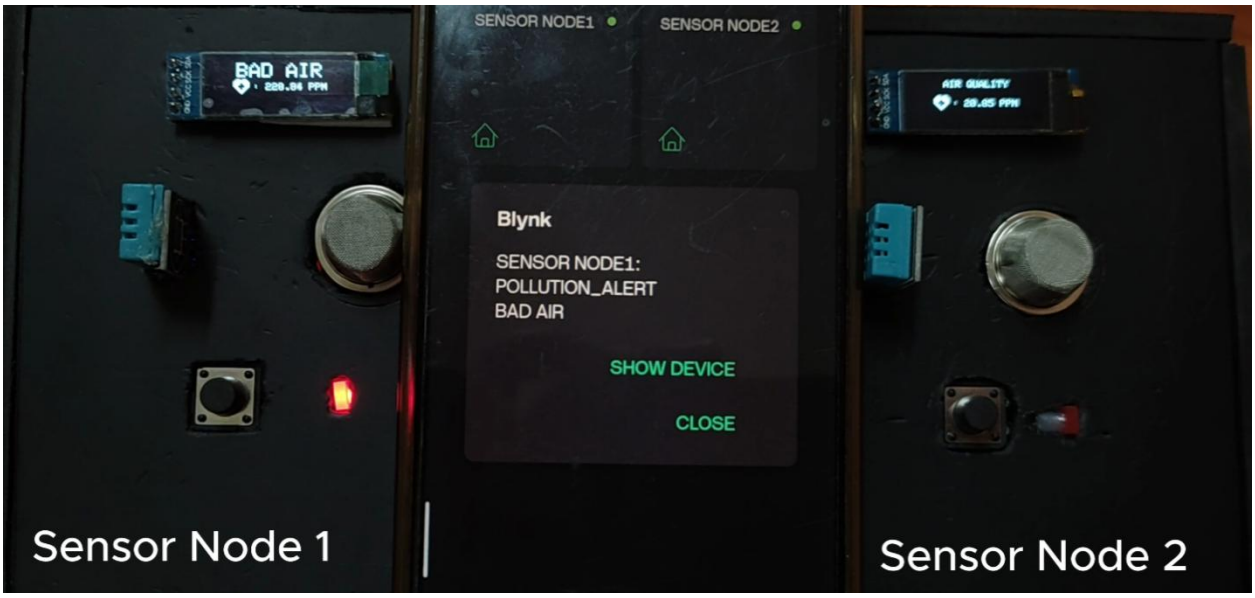


Fig. 4.6. Image of monitoring warning at sensor node [1]

III. Design Analysis

1. Checking delay

- **Display and alerts**

After conducting tests, it has been confirmed that displaying sensor values on both the Sensor Node and the Blynk Server occurs almost simultaneously and without significant delays. Sensor values are updated and displayed in real-time on the OLED screen of the ESP8266 and are also accessed and displayed seamlessly on the Blynk mobile app and web interface. This allows users to easily observe and monitor sensor readings without experiencing significant delays or asynchronous behavior between different platforms.

Simultaneously, the alerts are also sent to the phone almost immediately when the values exceed the threshold.

- **Button delay for debouncing**

As for the buttons on the Sensor Node, the programming includes debouncing to prevent button bouncing. However, sometimes delays in the code may cause interruptions in the program, preventing the button from accepting the OLED display screen transition.

SOLUTION: The team attempted to address this by using a timer to mitigate conflicts for the button press. However, this did not significantly improve the situation. Therefore, the best solution proposed was to reduce the button press delay, which proved to be acceptable.

2. The accuracy of the sensors

Since the project only implemented two sensors, the DHT11 and MQ-135, the team referred to comparison tables of sensor accuracy in various reference documents. An example is the comparison of temperature measurement accuracy between the DHT11 sensor and other sensors **[3]**:

SETPOINT	SENSOR STD	DHT11		DHT22		DS18B20	
		Value	Correction	Value	Correction	Value	Correction
20 °C	20.033	19.8	0.23	19.8	0.23	19.4	0.66
	20.059	19.8	0.26	19.7	0.36	19.3	0.75
	19.977	19.8	0.18	19.5	0.48	19.3	0.67
	19.989	19.8	0.19	19.6	0.39	19.4	0.62
	20.019	19.8	0.22	19.7	0.32	19.4	0.58
Mean	**20.015**	**19.8**	**0.22**	**19.66**	**0.36**	**19.4**	**0.66**
30 °C	29.999	29.8	0.20	29.5	0.50	29.3	0.69
	30.003	29.8	0.20	29.5	0.50	29.3	0.69
	30.013	29.8	0.21	29.5	0.51	29.3	0.70
	30.091	30.2	−0.11	29.7	0.39	29.5	0.59
	30.104	30.2	−0.10	29.7	0.40	29.5	0.60
Mean	**30.042**	**29.96**	**0.08**	**29.58**	**0.46**	**29.4**	**0.66**
40 °C	40.053	41.6	−1.55	39.4	0.65	39.4	0.67
	40.055	41.6	−1.55	39.4	0.66	39.4	0.67
	40.057	41.6	−1.54	39.4	0.66	39.4	0.68
	40.060	41.6	−1.54	39.4	0.66	39.4	0.68
	40.063	41.6	−1.54	39.4	0.66	39.4	0.68
Mean	**40.058**	**41.6**	**−1.54**	**39.4**	**0.66**	**39.4**	**0.68**

Table. 4.1. comparing the three sensors DHT11, DHT22, and DS18B20

From the image, it can be observed that among the mid-range sensor array, the DHT11 has an approximate fluctuation error of 1 degree Celsius. According to the test values in the temperature range of 20-40 degrees Celsius (room temperature), the DHT11 displays temperature quite accurately.

For the MQ-135 sensor, we have a comparison table with the MQ-137, a sensor module that costs more than 10 times the price of the MQ-135 and offers significantly better sensing capabilities **[4]**:

Day No.	MQ-135 (ppm)	MQ-137 (ppm)
1	29.15	31.89
2	28.38	41.05
3	29.5	29.64
4	29.43	36
5	33.128	42.15
6	28.25	30
7	29.1	31
AVERAGE	**28.95**	**33.6**

Table. 4.2. comparing the MQ-135 and MQ-137

The image shows that the MQ-135 still performs very well in measuring the PPM concentration of air. Although it has a relatively large error margin when compared to the MQ-137, it is still quite effective.

However, for the MQ-135 sensor, if it has not been powered for a prolonged period (around 1-2 days), the PPM values will spike when power is restored, requiring time to stabilize. Additionally, if the ESP8266 is not updated regularly, the MQ-135 may exhibit continuous fluctuations (random inaccurate values).

SOLUTION: Regularly update the software and power the MQ-135 for about 1-2 minutes to allow the sensor to operate normally.

3. Sensor range testing

A test was conducted to check the detection range of the MQ-135 sensor using gas from a lighter. The test involved releasing gas at distances ranging from 0 to 12 cm to measure the PPM (parts per million) values detected by the sensor. The results are presented in the table below:

DISTANCE (cm)	AIR QUALITY VALUE OF SENSOR NODE 1 (PPM)	AIR QUALITY VALUE OF SENSOR NODE 2 (PPM)
Initially (not gas)	0.48	3.4
2	>150	>150
4	>150	>150
6	>150	>150
8	60.78	70.78
10	12.5	19.87
12	3.5	1.8

Table. 4.3. MQ-135 range sense testing *1*

The PPM concentration represents the gas concentration, and from the data table, it is evident that the MQ-135 sensor detects gas concentration most effectively at distances under 10 cm, which is relatively short. However, since the amount of gas released from the lighter is quite small, the gas disperses into the air at greater distances, making it harder for the sensor to detect. Therefore, the gas detection range of the sensor is not highly rated but still acceptable.

SOLUTION: To achieve high efficiency and timely detection, the Sensor Node should be placed close to areas where gas leaks or explosions are likely to occur to ensure prompt detection.

IV. Result Analysis

Delay: The overall operation of the entire sensor node and system has been very stable, with a response speed that is almost instantaneous. However, the button exhibits a slight delay which does not significantly impact the system's performance.

Accuracy: The MQ-135 sensor needs a brief powering period to stabilize its readings. Despite this, the system consistently provides reliable and accurate measurements of air quality. Regular software updates help maintain this accuracy and address any minor issues that arise.

Sensor Range: Due to the relatively short sensing range of the sensors, it is advisable to place the Sensor Node close to potential gas leak sources to ensure accurate and timely detection of hazardous gases. This strategic placement enhances the system's effectiveness in monitoring air quality and providing prompt alerts.

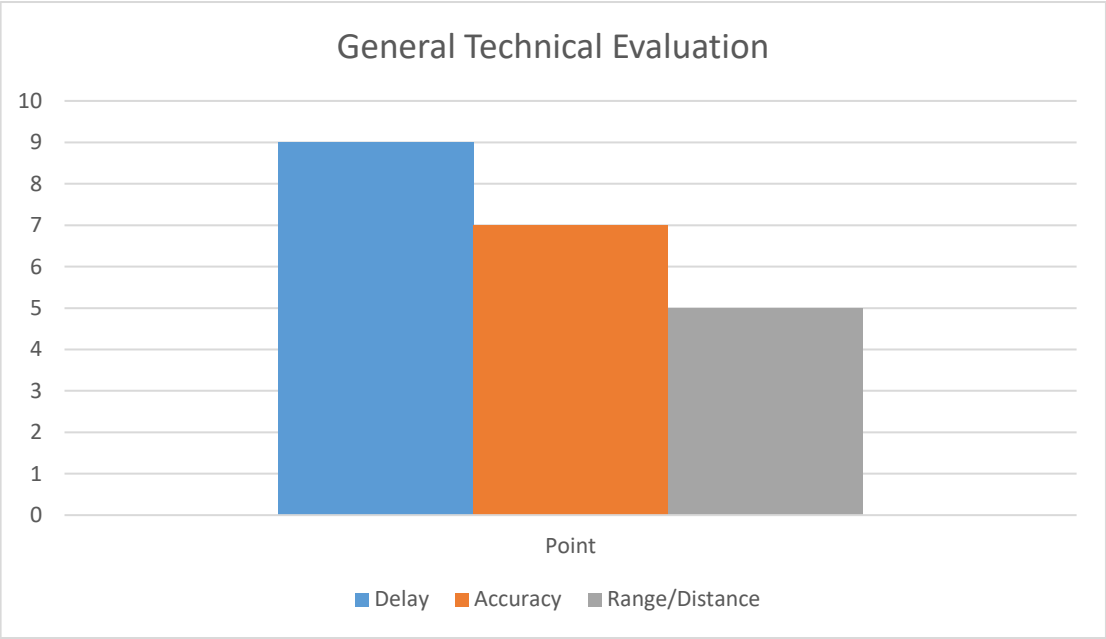


Fig. 4.7. Chart for evaluating technical *1*

Overall, the system performs reliably and effectively, ensuring real-time monitoring and alerts for air quality.

CHAPTER 5: RESULTS

The Air Quality Monitoring System using ESP8266 and Blynk Server successfully measures key air quality parameters including temperature, humidity, and various gas concentrations.

Fig. 5.1. The final image of the sensor node from the project [1]

Advantages:

- **Low Cost:** The system is affordable, making it accessible for various users.
- **User-Friendly Interface:** Easy to use, with monitoring available on both mobile devices and computers.
- **Real-Time Alerts:** Sends notifications when air quality deteriorates, ensuring timely awareness.
- **Real-Time Sensor Monitoring:** Allows immediate observation of sensor data changes at the sensor node.

- **Fast response time:** The system exhibits rapid response times, ensuring timely detection and alerts for changes in air quality.
- **Acceptable accuracy:** The sensors provide reasonably accurate measurements, enabling effective monitoring of air quality and gas concentrations.

Disadvantages:

- **Wi-Fi Dependency:** Requires a Wi-Fi connection to function, limiting its use to areas with network availability.
- **Indoor Use Only:** Designed primarily for indoor environments.
- **Software Updates:** Occasionally requires firmware updates for the sensor nodes to maintain reliability and avoid errors.
- **Short sensor range:** The sensors have a limited detection range, necessitating their placement in close proximity to potential sources of hazardous gases for optimal performance.

CHAPTER 6: CONCLUSION AND RECOMMENDATION

I. Conclusion of the entire project

Through the integration of advanced sensors, IoT technology, and data analytics, the Air Quality Monitoring System offers unparalleled accuracy and reliability in assessing air quality. This capability is particularly crucial in urban areas where pollution levels can fluctuate rapidly due to various factors such as traffic congestion, industrial emissions, and meteorological conditions.

Furthermore, the accessibility of this system empowers citizens to actively engage in monitoring their local air quality, fostering awareness and advocacy for cleaner air. By providing transparent and easily understandable information, the system encourages collective action towards reducing emissions and improving air quality standards.

Looking forward, the Air Quality Monitoring System will continue to evolve in response to emerging technologies and environmental priorities. Integration with smart city infrastructure, the Internet of Things (IoT), and machine learning algorithms holds the promise of enhancing the system's capabilities for predictive modeling, early warning systems, and targeted interventions in areas most affected by poor air quality.

In essence, the Air Quality Monitoring System serves as a beacon of hope in the quest for cleaner, healthier air for all. By leveraging technological innovation, scientific expertise, and community engagement, we can work together to mitigate the impacts of air pollution, protect vulnerable populations, and create a more sustainable future for our planet.

II. Suggestions for future project development

In the future, this project may be replaced recent sensors from better sensors for high performance. Thanks to that, a state-of-the-art product is manufactured. Moreover, we should build a unique MQTT system server to prevent product from interruption.

REFERENCES

[1] Somphop Chanthakit & Choopan Rattanapoka, "MQTT Based Air Quality Monitoring System using NodeMCU and Node-RED", *Seventh ICT International Student Project Conference* 2018.

[2] S. A. Z. Murad, F.A. Bakar, A. Azizan & M.A.M Shukri, "Design of Internet of Things Based Air Pollution Monitoring System Using ThingSpeak and Blynk Application", *Journal of Physics: Conference Series*, 2021.

[3] David Yulizar, Santoso Soekirno, Naufal Ananda, Muhammad Agung Prabowo, Ilham Fajar Putra Perdana, and Diar Aofany, "Performance Analysis Comparison of DHT11, DHT22 andDS18B20 as Temperature Measurement", *ICSES* 2022.

[4] Prachi Pahare, Preet Jain, "Study of reliability of electrochemical sensor MQ-135 and MQ-137 in concern of ammonia gas in male urinals implemented on IOT based system", *IJREAMI,* Vol-05, July 2019.

APPENDIX A

GUIDE SETTING UP BLYNK SERVER

To facilitate the initial programming, set up the sensor values and events to be sent to the Blynk Server beforehand.

Step 1: After accessing the **blynk.cloud** website, create an account and log in. In the **'Developer Zone'** , select **'+ New Template'**.

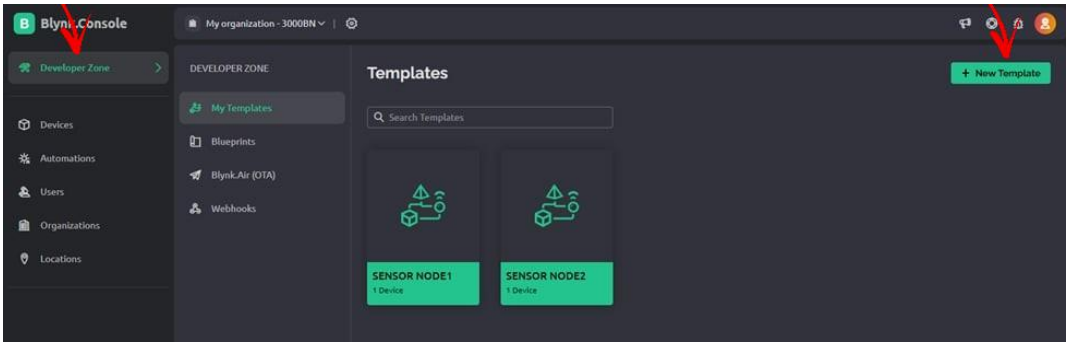


Fig. *Step 1 setting up for Blynk.*

Step 2: Create a template for your Sensor Node, **select ESP8266**, and **choose the Wifi connection.**

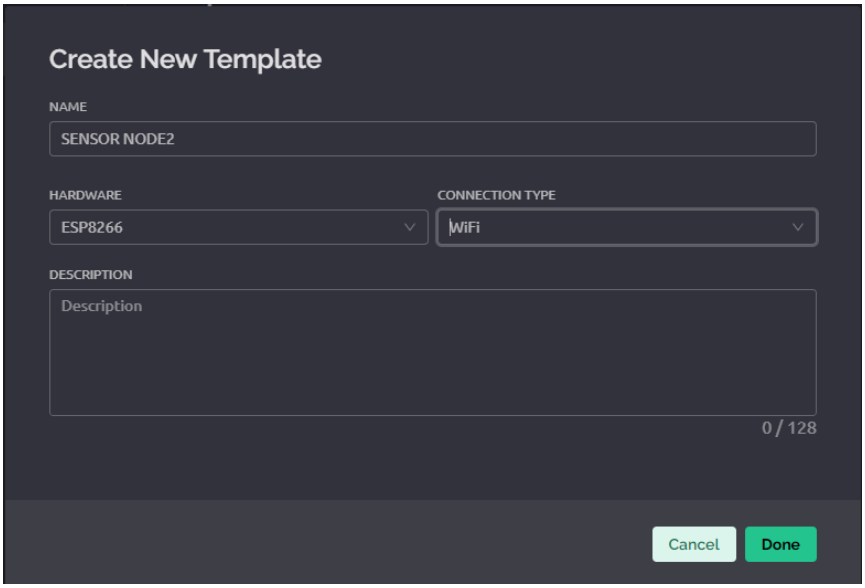


Fig. *Step 2 setting up for Blynk.*

Step 3: After creating the Template, under **the Datastream section**, **create variables for the sensors** as shown in the image below.

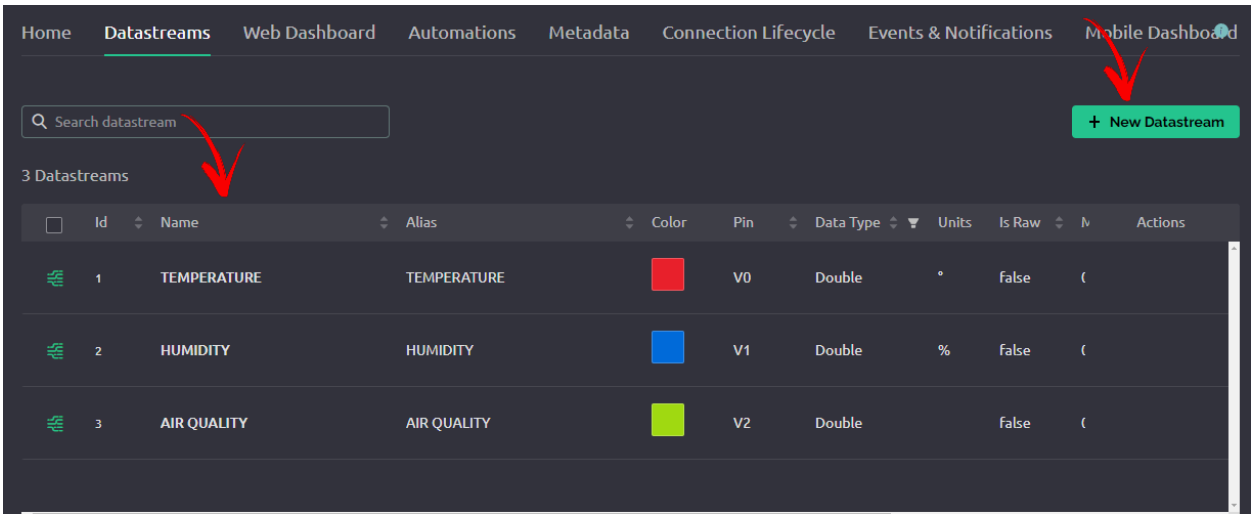


Fig. *Step 3 setting up for Blynk.*

Step 4: The Web Dashboard is where you design the sensor monitoring interface, you can fetch devices from **the Widget Box**.

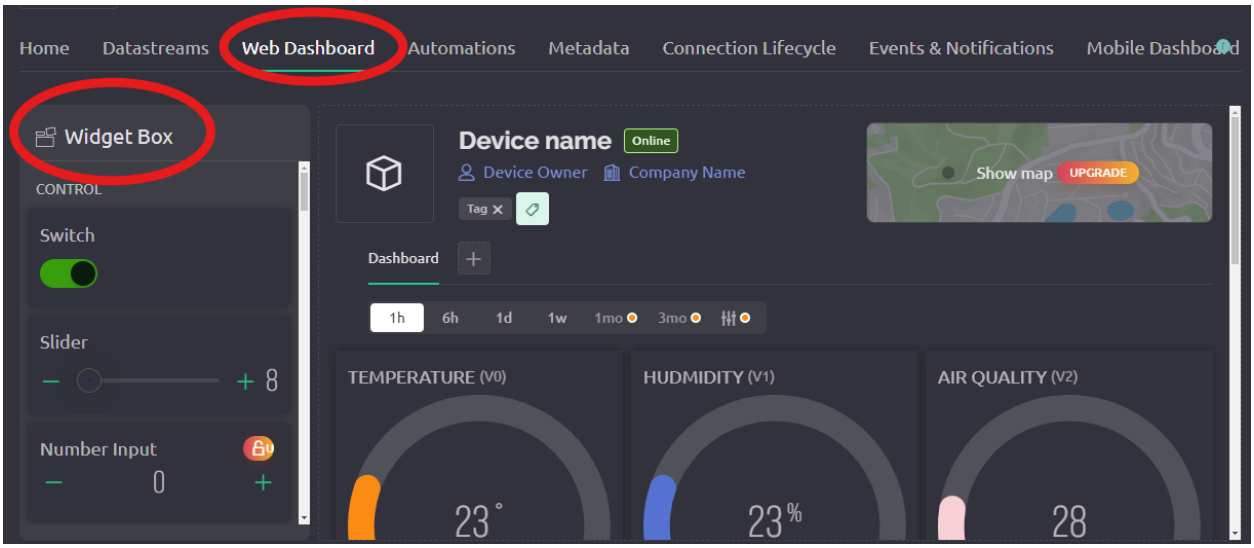


Fig. *Step 4 setting up for Blynk.*

Note: Remember to **click on the icon** and **select the Datastream values** you've created and wish to monitor.

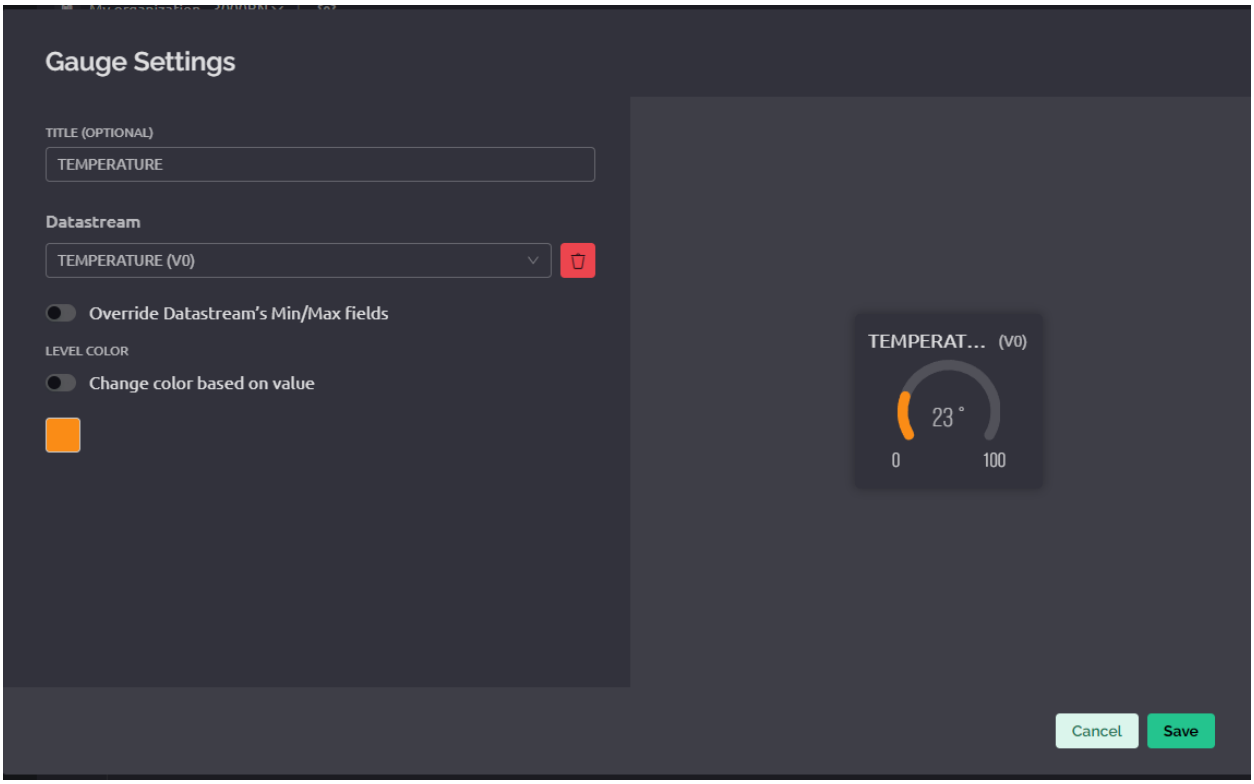


Fig. *Note setting up for Blynk.*

Step 5: If you wish to **stop at monitoring**, **step 6 is not necessary**. Use **'Events and Notifications' to create an alert event for your computer and phone.**

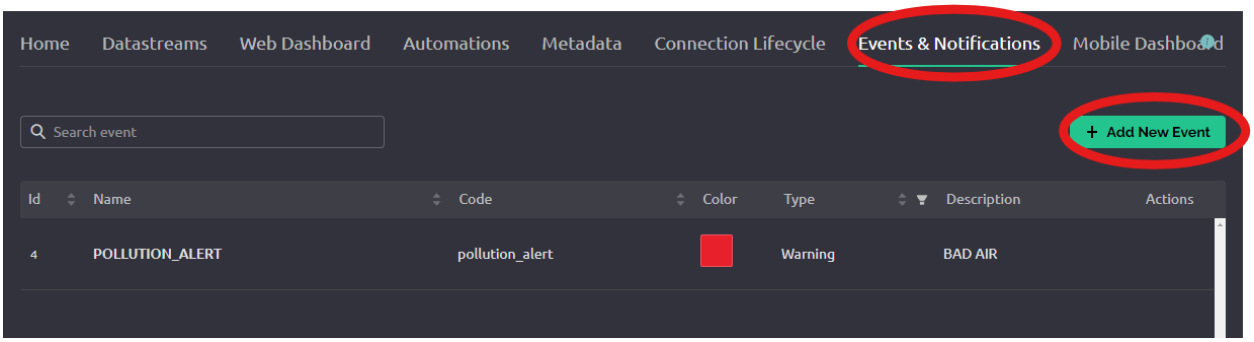


Fig. *Step 5 setting up for Blynk.*

Then create an event as shown in the settings below.

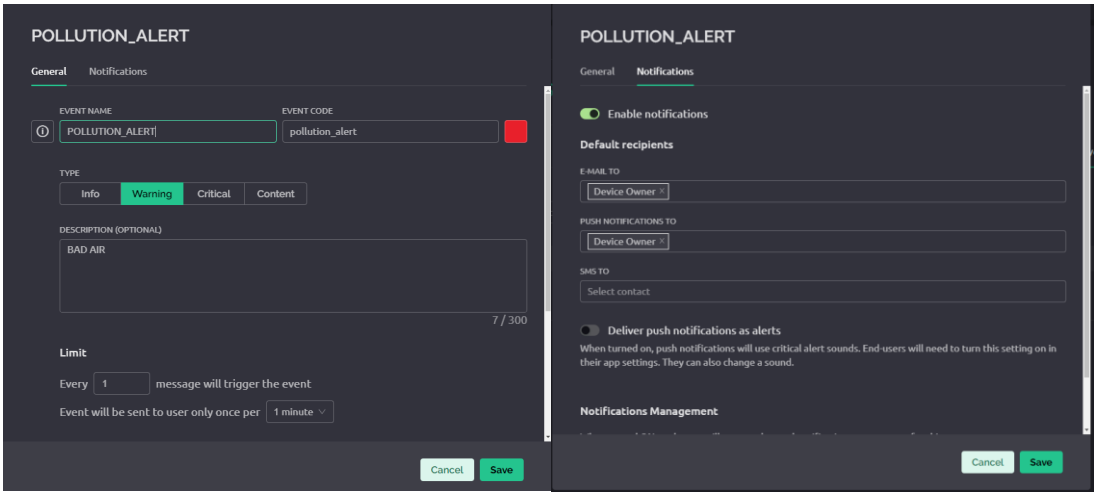


Fig. *Step 5 setting up for Blynk.*

Step 6: After finishing the setup, **click 'Save'** and **switch to the 'Devices' tab**, then **select your created Template.**

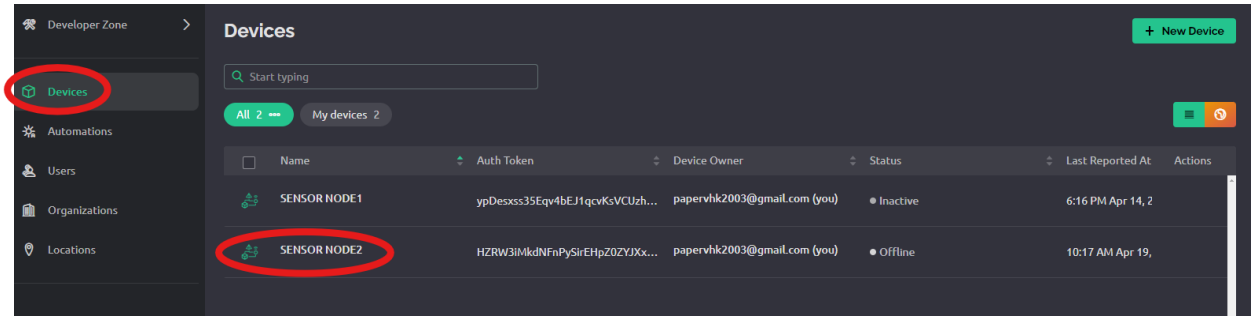


Fig. *Step 6 setting up for Blynk.*

Click on **'Developer Tools':**

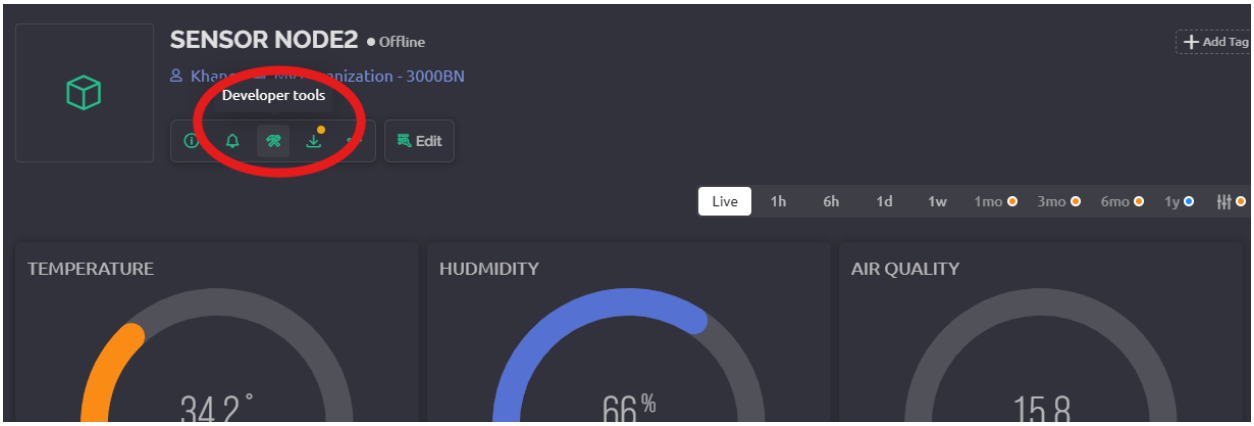


Fig. *Step 6 setting up for Blynk.*

The Firmware info will be the **Token for your program**. Each created Template will have a different Token.

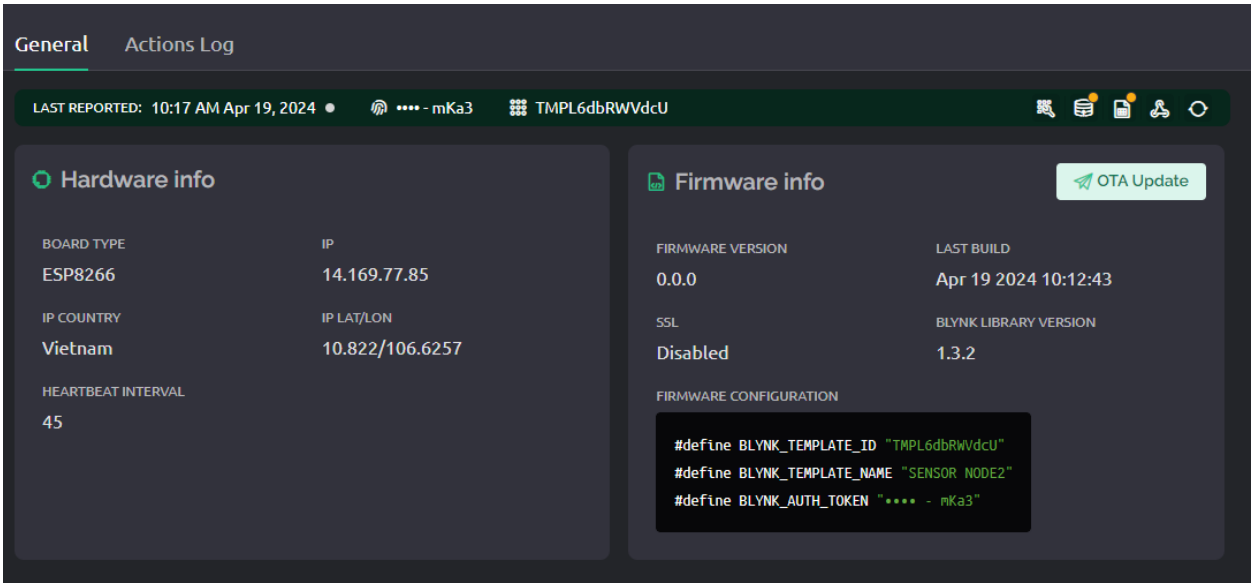


Fig. *Step 6 setting up for Blynk.*

Note: If you want to monitor from your phone, **download the Blynk IOT app** and **log in with the account you created** on the Blynk Cloud website. After logging in, the Template you created will be displayed. Now, you just need to **set it up similarly to step 4.**

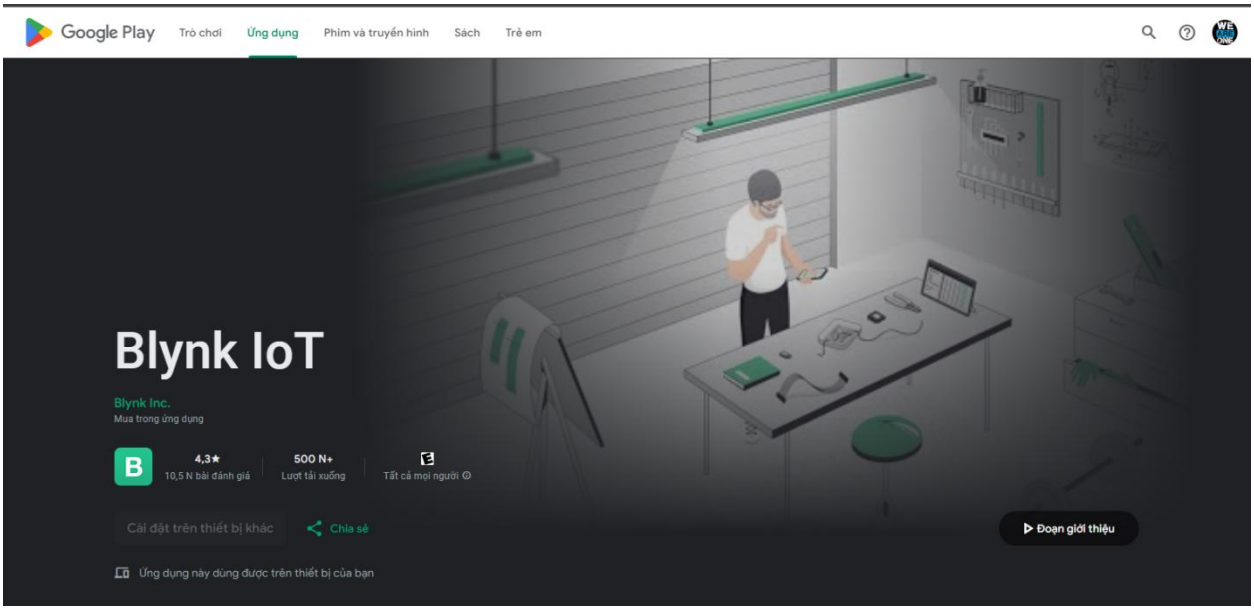


Fig. *Note setting up for Blynk.*

APPENDIX B

CODE AND EXPLANATION

The required libraries and explanation of code programming

The required libraries are as follows:

No.	Content	Note
1	**Provides Wi-Fi connectivity for ESP8266 modules.**	**#include ESP8266WiFi.h**
2	**Graphics library for drawing on displays and controls SSD1306 OLED displays.**	**#include <Adafruit_GFX.h>** **#include<Adafruit_SSD1306.h>**
3	**I2C communication library.**	**#include <Wire.h>**
4	**Reads DHT temperature and humidity sensors.**	**#include <DHT.h>**
5	**Reads MQ135 gas sensor data.**	**#include "MQ135.h"**

All code and explanation:

The programming code for both Sensor Node 1 and Sensor Node 2 is available at the GitHub link below, along with explanations. The code allows button presses to change the OLED display of sensor values, and it also connects to Wi-Fi to report threshold alerts to Blynk.

GitHub Link: https://github.com/GauTapCode/AirQualityMonitoring/tree/main